Soil vs. Lunar Regolith

Arduino Moisture Sensor Experiment — Student Worksheet

Name: **Date:**

Part 1 — Background

What is lunar regolith? How does it differ from soil on Earth?

Why might growing plants on the Moon be challenging? List at least two reasons.

Part 2 — Initial Hypothesis (complete before handling the materials)

Which material do you predict will retain more moisture after 100 mL of water is added? Circle your prediction.

○ Potting soil ○ Regolith simulant ○ About the same

Explain your reasoning. What do you already know about these materials that led to your prediction?

Part 3 — Pre-Experiment Observations (before adding water)

Examine each material and record what you notice below. Use your senses (except taste).

Property	Potting Soil	Regolith Simulant
Color		
Texture		
Particle size		
Smell / other		

Based on these observations, which material do you now think will absorb water more quickly? Did your observations change your thinking from Part 2?

Part 4 — Revised Hypothesis (after observations, before adding water)

Note: *Now that you have observed the materials, you may want to update your prediction.*

Does your prediction change after observing the materials? Circle one.

○ My prediction stays the same ○ I am changing my prediction

If you changed your prediction, what did you observe that changed your thinking? If you kept it the same, explain what reinforced your original reasoning.

Part 5 — Data Collection

Record your moisture sensor readings (in Volts) for each condition. Take 3 readings per cup, then calculate group and class averages.

	Potting Soil (V)	Regolith Simulant (V)
Trial 1		
Trial 2		
Trial 3		
Group average		
Class average		

Sensor dry value (calibration): _____ V Sensor wet value (calibration): _____ V

Part 6 — Analysis

1. Which material retained more moisture? By how much? Use specific values from your data.

2. Did your final results match your revised hypothesis from Part 4? What was the same or different?

3. Look at the class data on the board. Were all groups' results similar? If there was variation between groups, what might explain it?

4. Connect your findings to the physical properties you recorded in Part 3. Why do you think the materials behaved differently?

Part 7 — Conclusion

Use claim-evidence-reasoning (CER): *State your claim, support it with specific data, and explain the connection.*

What effect does soil type have on moisture retention, and what does this mean for growing plants?

Part 8 — What If?

Answer at least two of the following. Use evidence from today's experiment.

What if you mixed potting soil and regolith together?

Predict the moisture reading for the mixture. Would it be closer to the soil result, the regolith result, or somewhere in between? Why?

What if you doubled the amount of water added?

Would the difference between conditions grow, shrink, or stay the same? Explain your reasoning.

What if you were an engineer designing a farming system on a Moon base?

Based on today's data, what would you need to add or change to make regolith support plant life?